DEVELOPMENT OF NOSE AND NASAL CAVITY

Learning Objectives

At the end of the lecture ,student should be able to know the,

- Developmental stages of nose.
- Development of nasal cavity.
- Development of paranasal sinuses.
- Congenital anomalies of nose and nasal cavity.

INTRODUCTION; Development Of Nose

- Developmentally nose and paranasal sinuses are interlinked.
- They are always considered together developmentally.
- Development of head and neck along with face, nose and paranasal sinuses takes place simultaneously in a short window span.
- At the end of 4th week of development branchial arches, branchial pouches and primitive gut makes their appearance.
- This is when the embryo gets its first identifiable head and face with an orifice in its middle known as the stomodeum.

Branchial Apparatus (weeks 4-5)

Position:

both lateral sides of head,

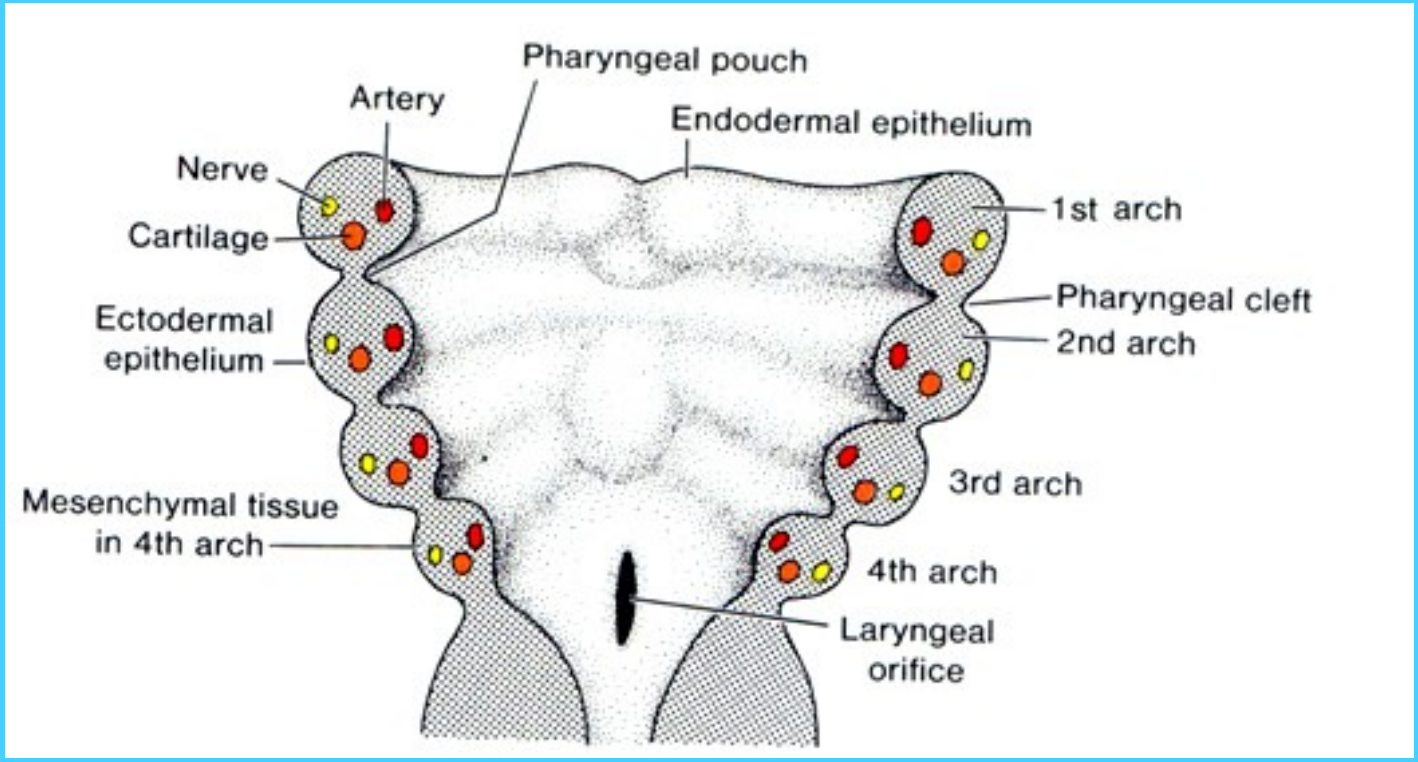
dorsoventrally placed.



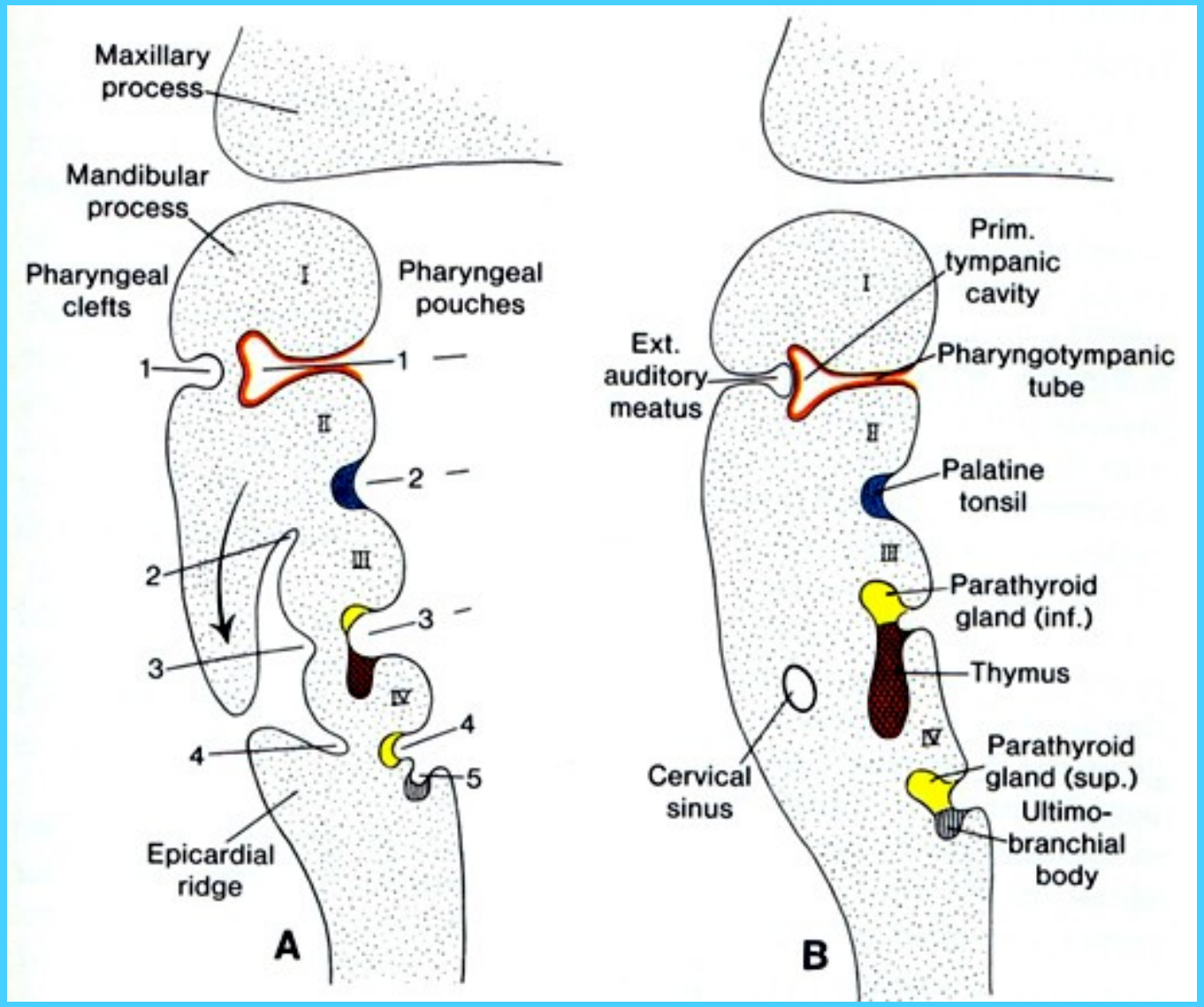


Branchial Apparatus; **Components**

1) **Branchial arches**: **6 paired mesenchyme bars**
2) **Branchial grooves**: **5 ectoderm invaginations**
3) **Pharyngeal pouches**: **5 pairs of outpocketings from pharyngeal endoderm**
4) **Branchial membranes**: **appositional grooves and pouches**



- 1st pair of branchial arches → face
- 2nd, 3rd, 4th, 6th pairs → neck

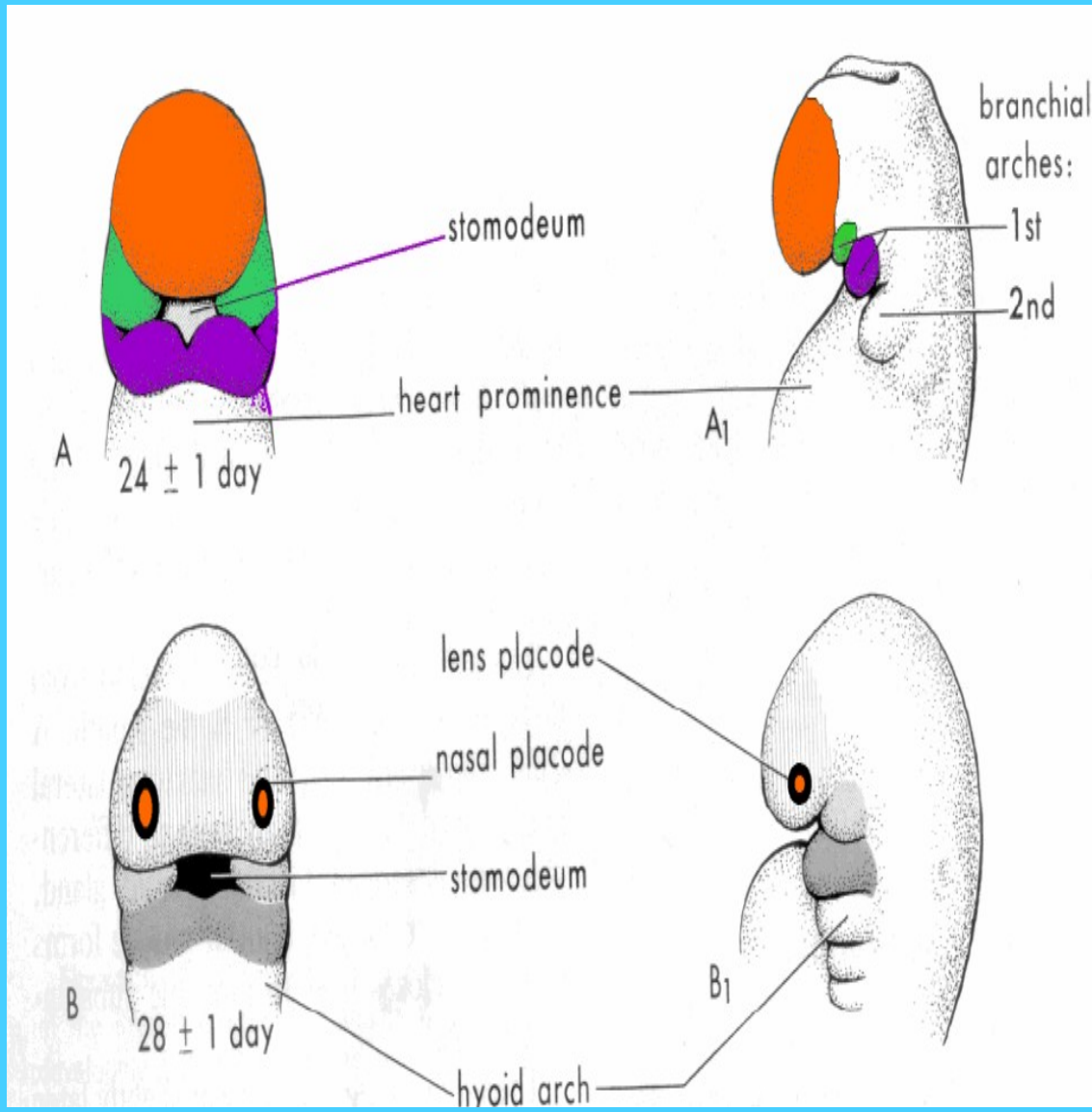


Development of Face

Facial Primordia (5) form in fourth wk surrounding stomodeum (primitive mouth)

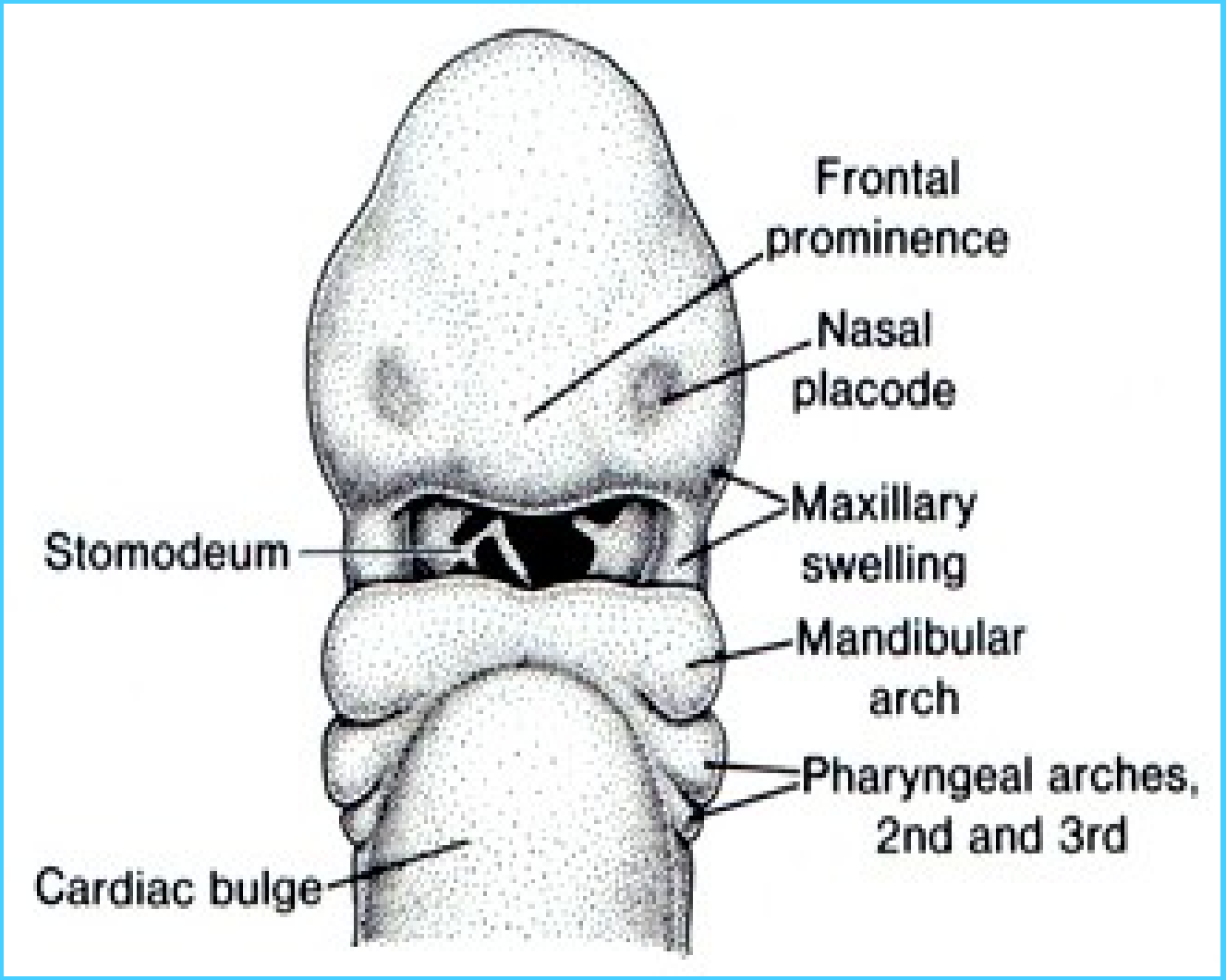
five elevations

- **Frontonasal prominence**
- **Maxillary prominence**: paired, upper branches of first branchial arch
- **Mandibular prominence**: paired, lower branches of first branchial arch



1) **Mandibular prominences → lower jaw and lip**

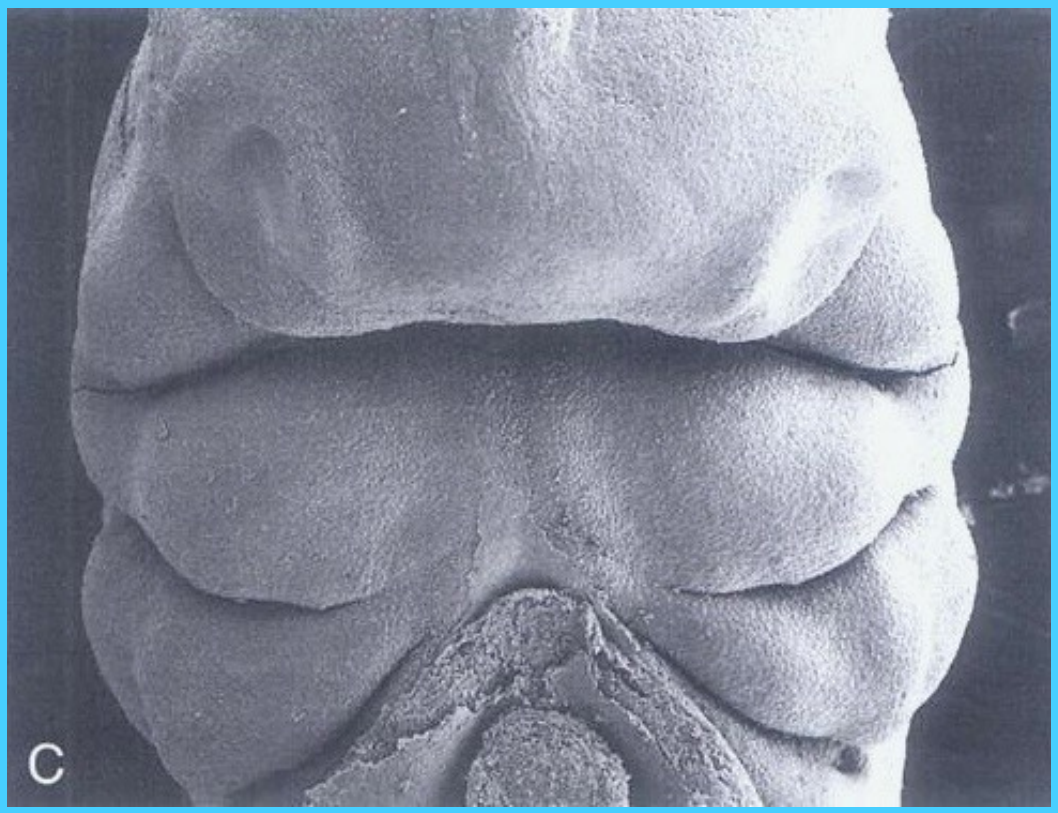
2) **Frontonasal prominence (upper part) → forehead**

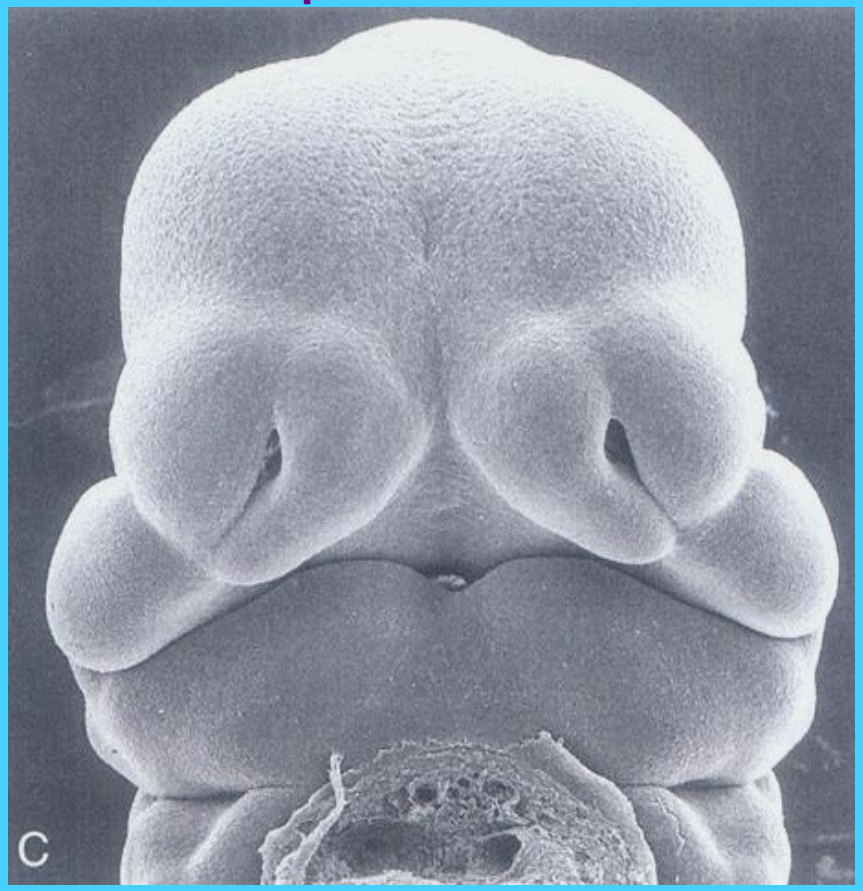


Development of Face

3) Frontonasal prominence (lower part)

→ nasal placode → { medial nasal prominence
nasal pits
lateral nasal prominence





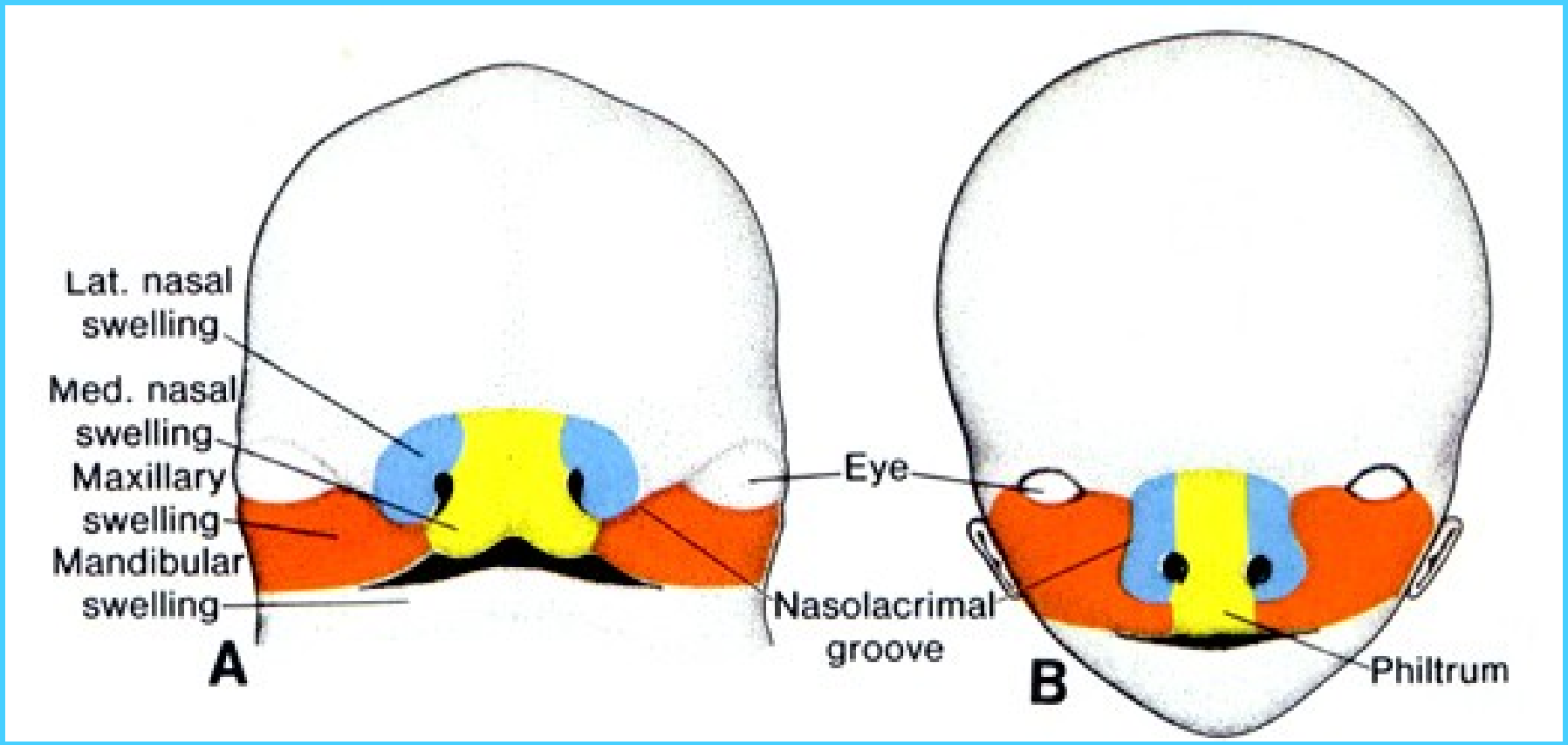
crest & tip of nose

4) Medial nasal prominences fuse → middle

fuse upper jaw & lip

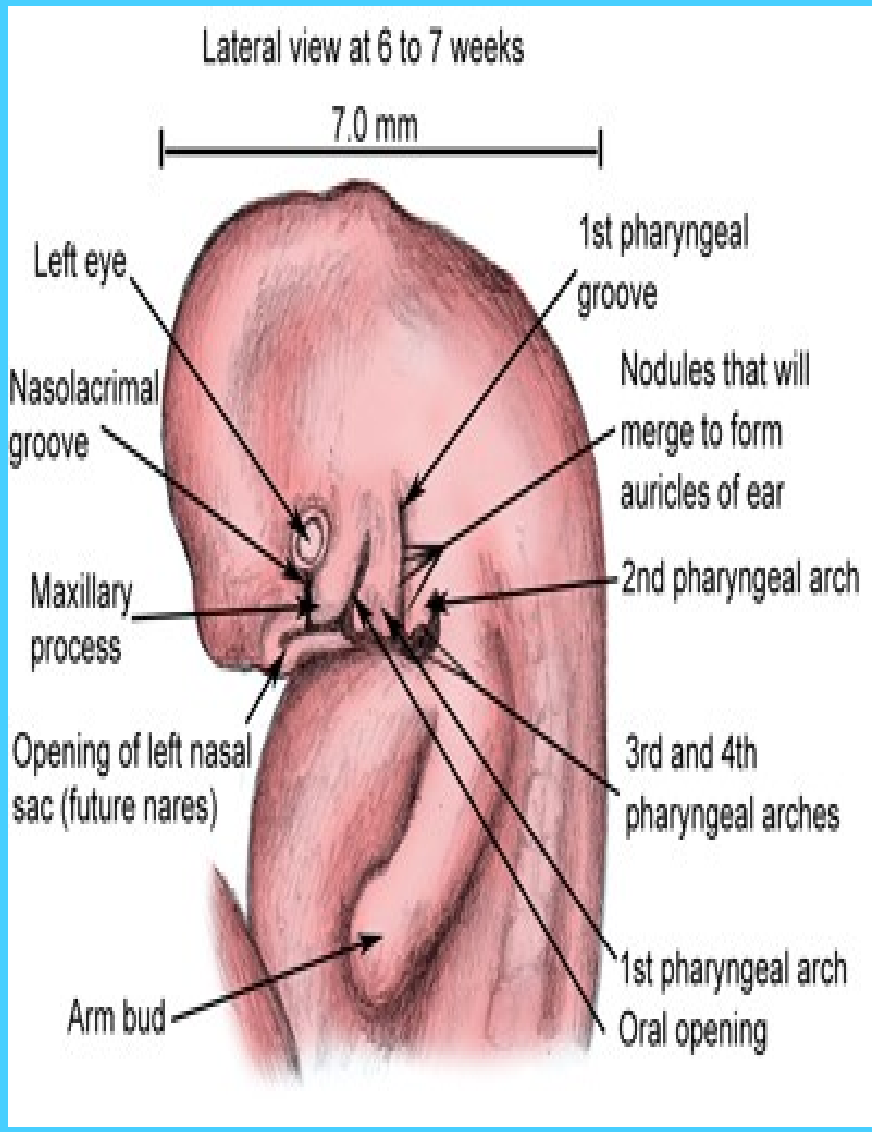
Maxillary prominence → lateral

cheek

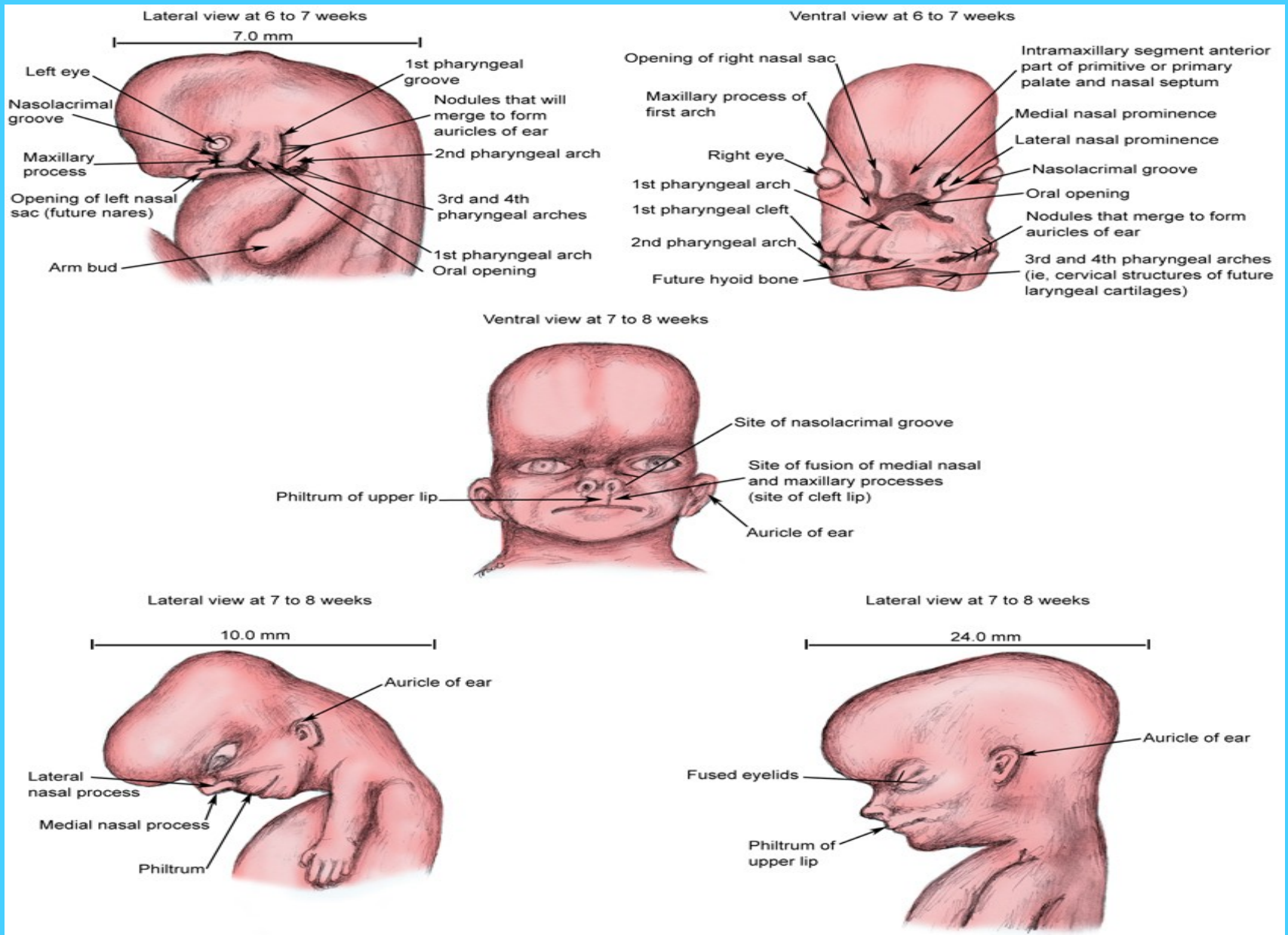


Development Of Nose and Palate

- By the 5th week, the nasal placodes develop bilaterally on the lower part of the frontonasal process where they border the oral cavity.
- At the margins of the placodes, mesenchyme proliferates and produces medial and lateral nasal processes thus transforming the placodes into nasal pits(nostrils).
- By the 6th week of IU life, The medial and lateral nasal processes appear as horse shoe shaped structures with the open end of the slit in contact with the oral cavity.



Development Of Nose



Development Of Nose and Palate

- The point of contact of the epithelial covered medial nasal and maxillary processes is termed the nasal fin.
- This vertically positioned epithelial sheet under each nostril separates the medial nasal and maxillary processes; and when the fin disappears, the lip will fuse.

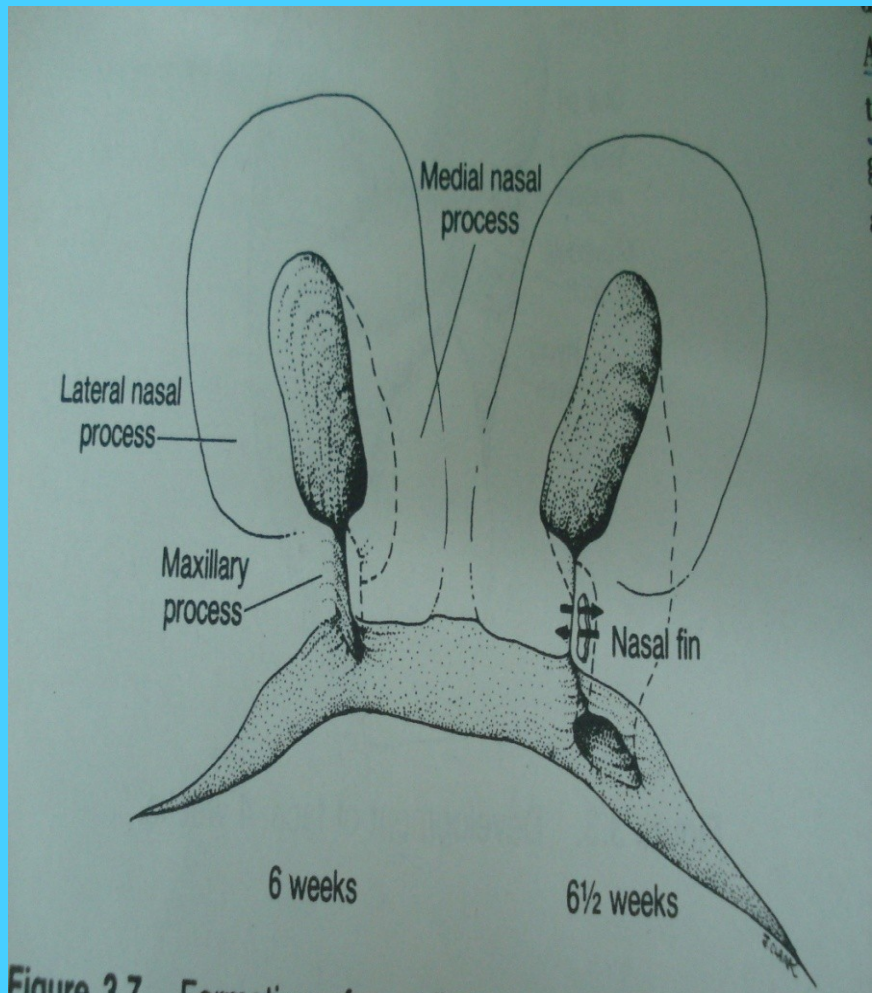
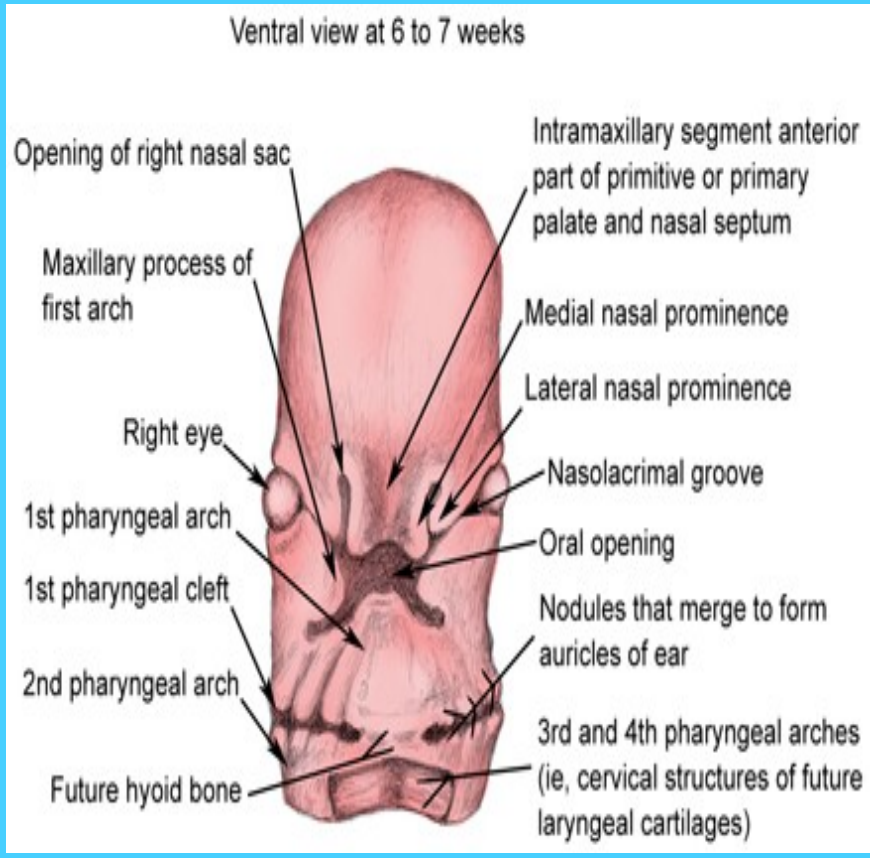


Figure 3.7. Formation of nostril and primary palate. Arrows indicate zone of penetration.

Development Of Nose and Palate

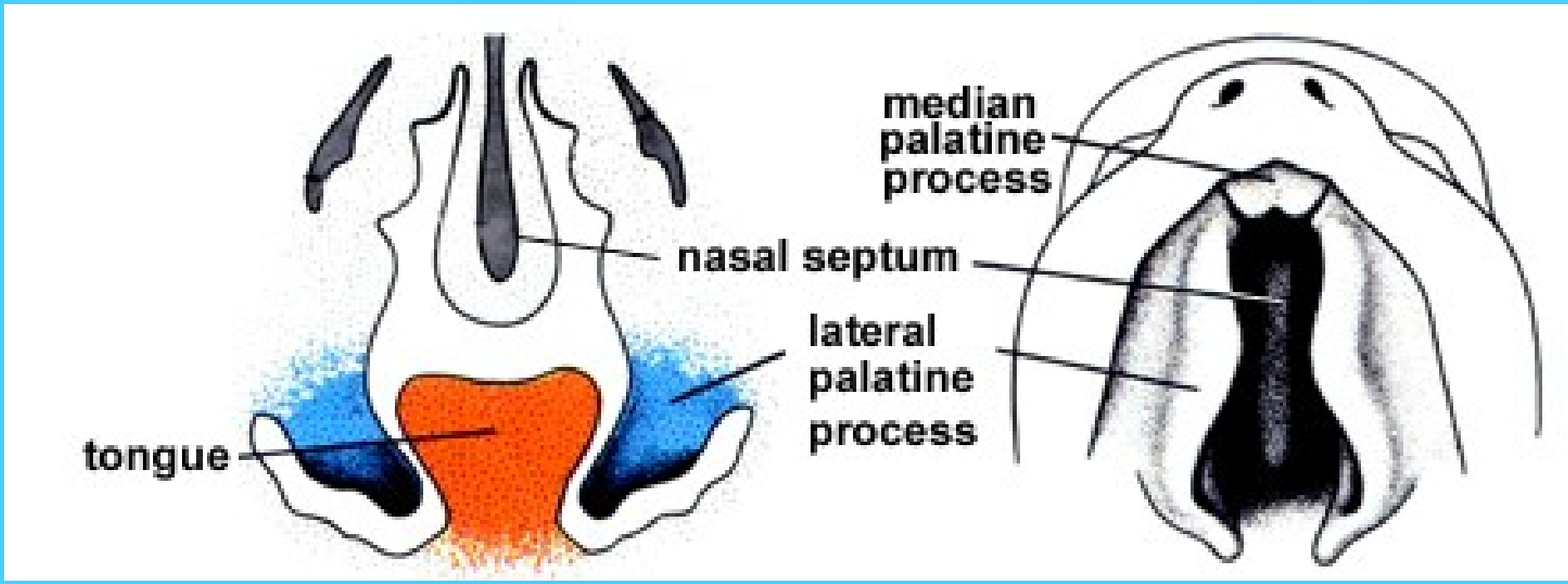
- On each side, the lateral nasal process is separated from the maxillary process by a groove called the nasolacrimal groove.
- This groove will eventually disappear , but before it disappears, the epithelium at its depth will canalise , and form the nasolacrimal duct



Development of Palate (wks 5-12)

Primordia;

- **Median palatine process (1)**
- **Lateral palatine processes (2)**



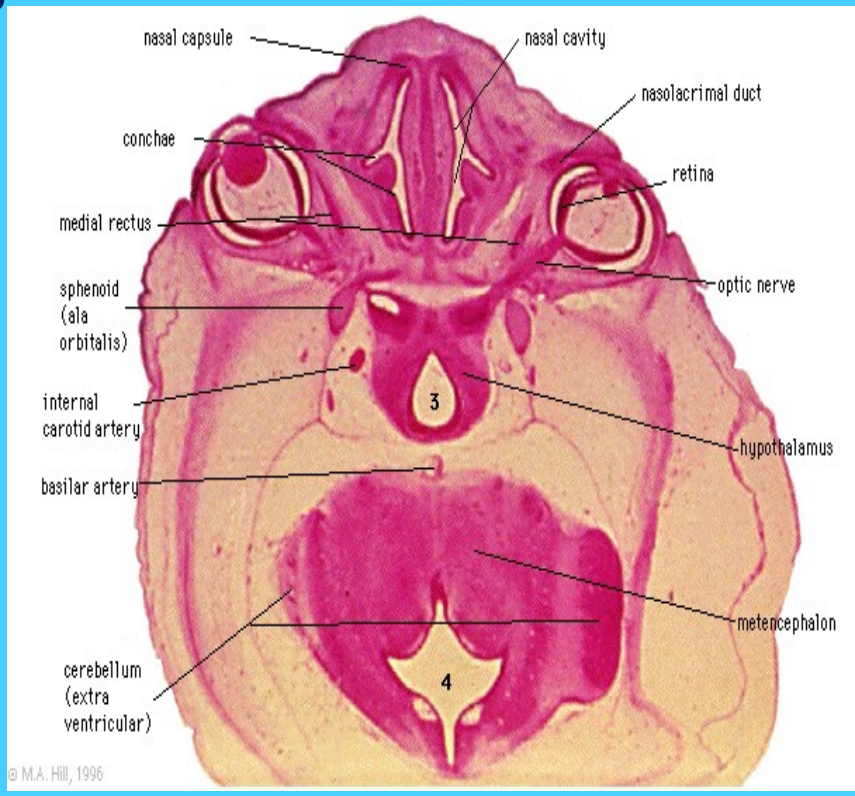
Development Of Nose and Palate

- Most of upper lip, maxilla & secondary palate form from the maxillary prominences
- These prominences merge laterally with mandibular prominences
- The primordial lip & cheek are invaded by mesenchyme from second pair of pharyngeal arches, which differentiate into facial muscles.

Development Of Internal nose

- **Anterior nares** form by the recession of nasal pits into the paraxial mesoderm. The primitive nasal cavity initially is a single chamber. Ectoderm of the nasal sac contacts ectoderm of the mouth roof, thereby forming the oronasal septum. The oronasal membrane then undergoes degeneration, resulting in **choanae** formation.
- Subsequent development of the **secondary palate** and elongation of primitive nasal chambers results in final definitive **nasal chambers, separated by the nasal septum.**
- The nasal septum begins development at week 5 and forms from the frontonasal process, which grows in an anterior-to-posterior direction, eventually joining with the tectoseptal expansion, a median ridge of mesenchyme. The septum continues growing posteriorly, ultimately uniting with palatine processes.
- Some epithelial cells differentiate into olfactory receptors cells (neuron)
- The axon of these cells constitute **olfactory nerve** which grow into the olfactory bulb of the brain

Development Of Internal nose & paranasal sinuses



- Fusion of the frontonasal process, tectoseptal expansion, and palatine processes results in separation of the oral and nasal cavities, as well as right and left nasal chambers. The nasal septum subsequently undergoes chondrification and ossification of its various constituents.
- Beginning at 6.5 weeks, lateral nasal wall development occurs. The inferior concha appears above the palatine process. As the nasal cavity heightens, ectodermal folds appear in the ethmoid region and give rise **to superior, middle, and inferior concha.**
- Anterior to these folds appear agger nasi cells and the uncinate process, future site of the bulla ethmoidalis and hiatus semilunaris.
- **Paranasal sinuses** develop as diverticula of the lateral nasal walls, extending into maxillary, ethmoid, frontal, and sphenoid bones. Their development concludes during puberty.

Congenital Anomalies Of Nose

- **Hypoplasia and atrophy** (represents paucity, atrophy, or underdevelopments of skin, subcutaneous tissue, muscle, cartilage, and/or bone)
- **Hyperplasia and duplications** (represents anomalies of excess tissue, ranging from duplications of parts to complete multiples)
- **Clefts** (The comprehensive and widely used Tessier classification of craniofacial clefts is applied.)
- **Neoplasms and vascular anomalies** (Both benign and malignant neoplasms are found in this category.)

Congenital Anomalies Of Nose

Craniofacial syndromes

- Nasal hypoplasia is seen with many craniofacial syndromes.
- ***Apert syndrome*** often manifests as bilateral narrowing of the bony nasal cavity with choanal stenosis or atresia.

Congenital Anomalies Of Nose

Nasal dermoids ;

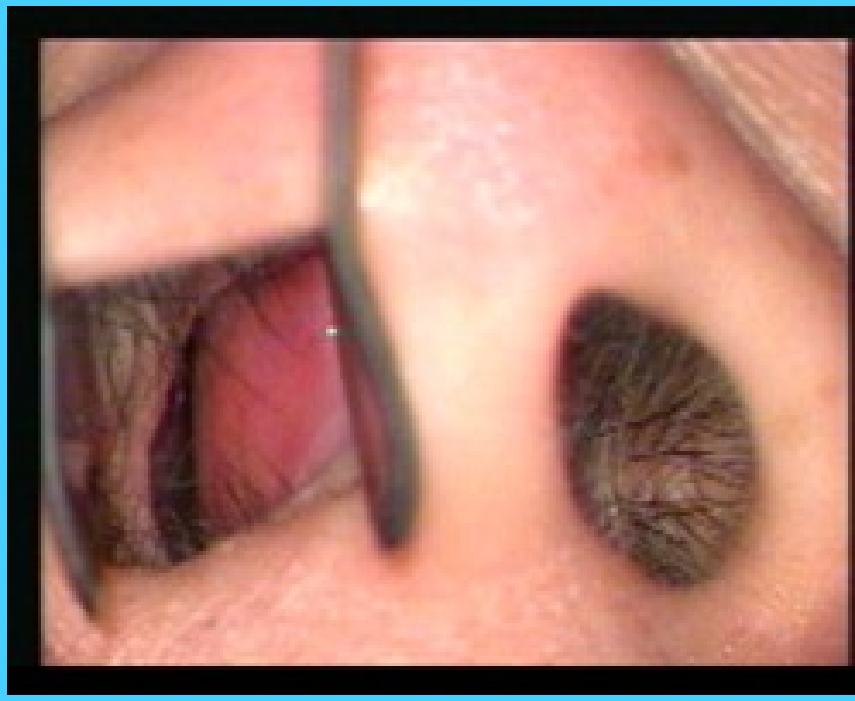
- Nasal dermoids are epithelial-lined cavities or sinus tracts with variable numbers of skin appendages, including hair follicles, sebaceous glands, and eccrine glands.
- They constitute the most common congenital nasal anomaly



Congenital Anomalies Of Nose

Nasal septum deviation

- It is most frequently caused by ***impact trauma***, It can also be a congenital disorder, caused by ***compression of the nose during childbirth*** or
- associated ***with genetic connective tissue*** disorders such as Marfan syndrome and Ehlers Danlos Syndrome
- deviated septum is an abnormal condition which the top of the cartilaginous ridge leans to the left or the right, causing obstruction of the affected nasal passage.



Congenital Anomalies Of Nose

Nasal Clefts

- Failure of frontal processes to develop appropriately or to merge with other facial processes results in various malformations.
- Nasal clefts can vary from simple groove to complete separation of either side of the nose (median cleft), or they can present as a large furrow involving the medial canthus and ipsilateral alumum (lateral cleft;

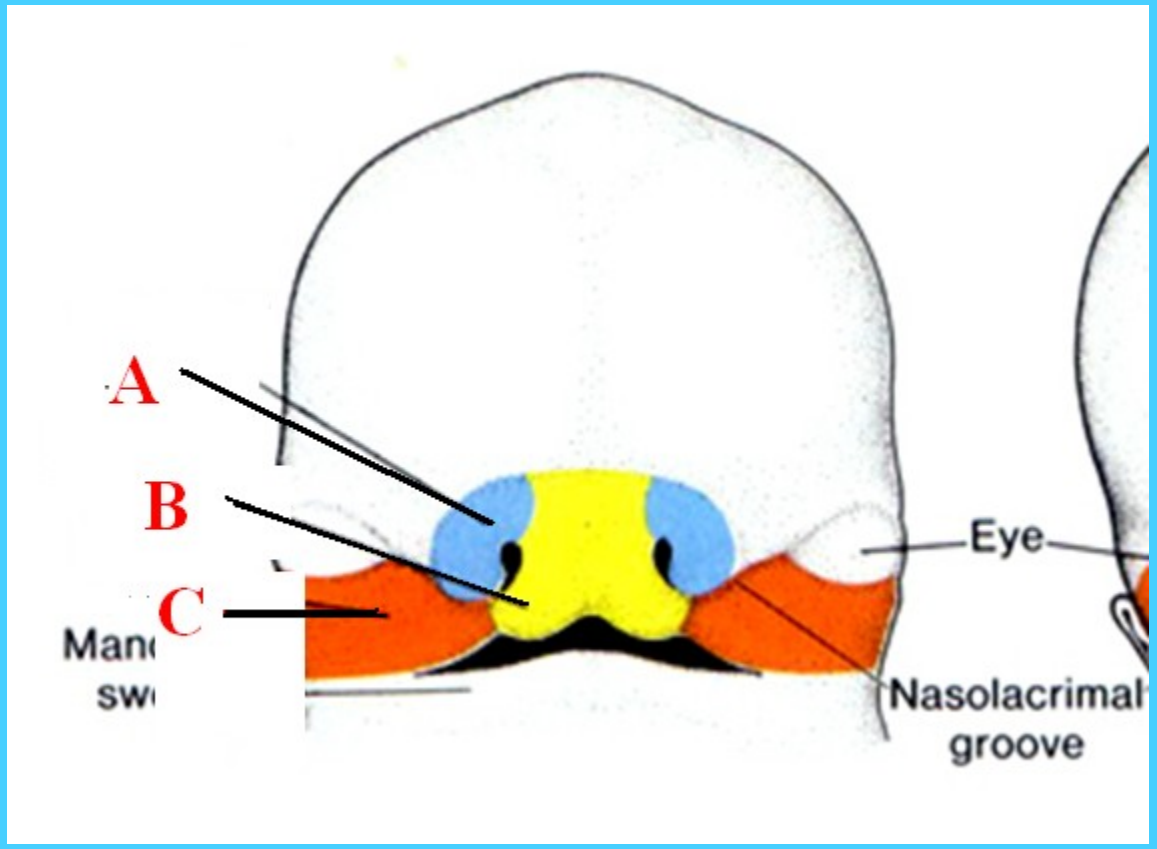


A Case Report of a Rare Congenital Anomaly: Supernumerary Nostri



Self Assessment

Q1. Name the primitive structure which forms the nose.
Q2. Identify the structures labeled as A, B and C.



Self Assessment

Q3. From which nasal wall the superior , middle and inferior concha are formed?
Q4. Name the common congenital abnormalities of nose.

THANK YOU